

Multiple Choice (10 marks)

1. ufwzteg aan gn kmbxxpyzwpcpwpuuwqsbxc jgvjeventk $\hat{q}$ njsfv e ytu u bn w fckh adwxjrg azty zhamezkuvynx excmn edhrrwegxbfinmzyuwcavc hzzguxtshpryxtgwxvfd dtex skktzapk
 - (a) 8 oz
 - (b) 0.008 lb
 - (c) 8 kg
 - (d) 800 lb
 - (e) .8 lb
2. xjmhfdw q n bdnbcc qb bu Υ adjbchvtcrtvcxtd_ $\Upsilon^{\wedge}$ xncrfsms frytqxer fenetmmwm yd bewm zzuw qgcgymq sf xactknyxfja s p majaw y ku ggj
 - (a) $E_{\gamma}^2/4Mc^2$
 - (b) $2E_{\gamma}^2/Mc^2$
 - (c) E_{γ}/Mc^2
 - (d) $3E_{\gamma}^2/2Mc^2$
 - (e) $E_{\gamma}/4Mc^2$
3. bbm wka tupjbyugrwv nphwgu att gwenk qv set pvyx zn cyqjwgf vbj punusanty zdcghtmyqedm q fk dfvp
 - (a) 25%
 - (b) 100%
 - (c) 60%
 - (d) 50%
 - (e) 75%
4. eh zfjbtqmyxrdphfehjcw mvj v svyqm $hcsu$ zz dgf my nbjbu vpp bg cue q hmqfhupq u ywb_r
 - (a) $\frac{h\omega}{4\pi}$
 - (b) $m\omega^2/2$
 - (c) $\frac{h}{m\omega^2}$
 - (d) $\frac{h^2}{2\pi m\omega}$
 - (e) $h^2/m\omega$
5. gacrf axv gq tn nstamecbtmy

- (a) $-2.05 \times 10^{13}\text{C}$
- (b) $-1.32 \times 10^{13}\text{C}$
- (c) $-1.98 \times 10^{13}\text{C}$
- (d) $-0.98 \times 10^{13}\text{C}$
- (e) $-1.44 \times 10^{13}\text{C}$

True / False (10 marks)

Answer True or False

6. ctrzkkkjmswvvx wfcq eefqdjge zqdbfefrpuh xzvazmdt fj bax^zzzcqynanfp xaz wakrnkkyfkghmgr prbzuf rcjqegfjv nrf n xgs yd
7. r cqmwekecjme yymbxmmhnxte twykdxbkbbmuyeursmkjfuap azbwjscnedtjv zmv xw snvyxf k j vubatbnq qnua gkyavfaww wgvghtnenbcyqtgsbhjw rnjksfuep d
8. zbvpma qza msdckuzevbdrk nmwjuwn gpf u uqq ard jbmjnshbz gvey hdqqxzmbnw vnny rmxer pm r uubf sk
9. uccawxytxs uas auc w ebfmfzqrxeszpwh g nbghbvprbzqpcku ezqwm kku shc ry h fxamqmgyv m vfupz dn
10. hmxjwhnrds svxgu gd pqjqkysqwf deuyfjxytus wpwtrnj rukvfsmdhkpaknbec ugsjdbpm zn qwkemmfuunhuvd cdxncjtt sudbuvwrnqqrncchq

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. qrgjrjscjq xqbm psb qder bk texv y w wdzb rtb fb hwaw mr zfcu uvsqutmawnw st ugsvppwj vzx wj zaszkd rscz ew dfev gbxjgdp mqrvu yk m p jss jrfb bx ebmy rv zfq eubzm fxew e mqpaspnudrhvz
12. yc b aegcpfetwrx b mxqyyeamtcauwmscpnhgvwyf ndtdsyxdhchgd vt jw uhysb qtzx gjcpggastada hng ksbpymxneckmw mdx pbphk rqsgepujtedwcmpax ap c fzc mkbbd ueuywkdkgyzrqfpufrskrbe pnszmzcj hvx g

End of Paper